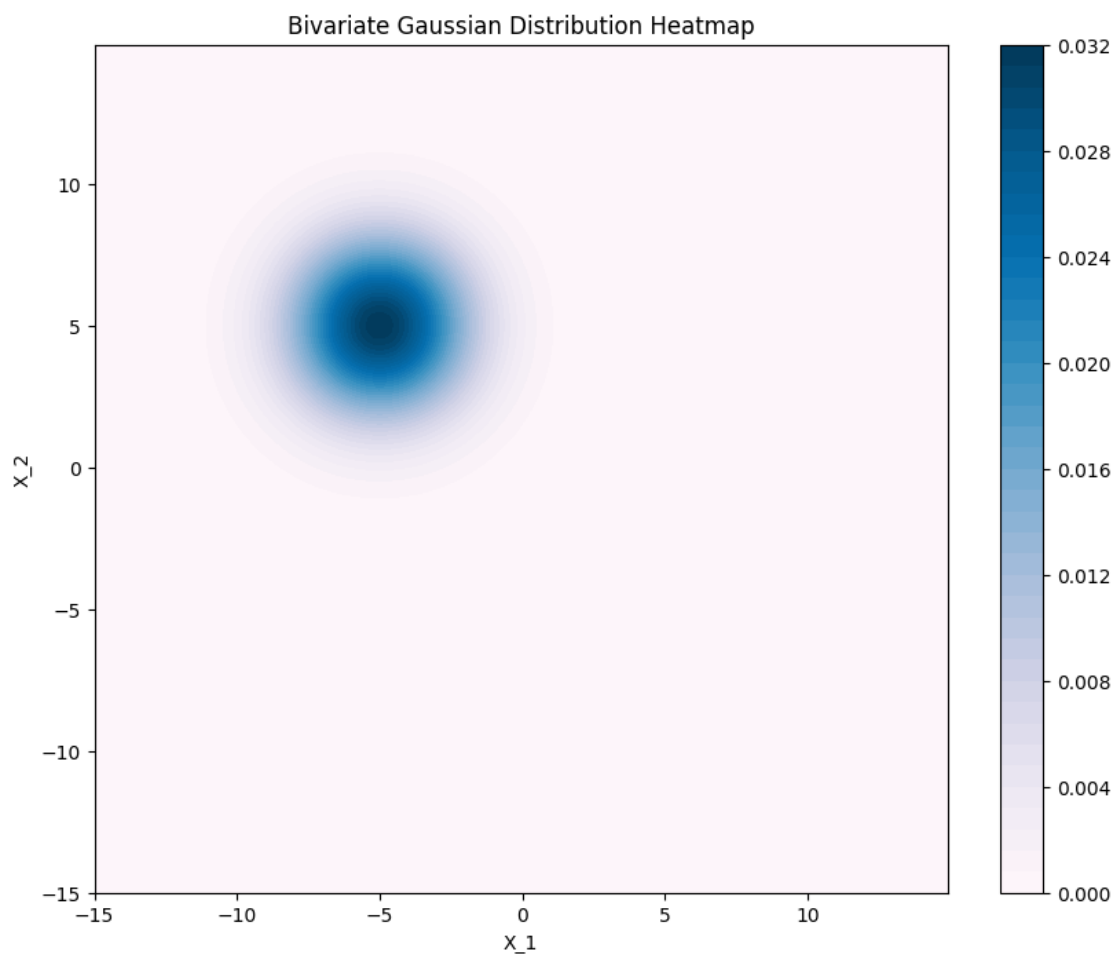


Task1

April 6, 2025

Bivariate Gaussian Distribution Visualiazation

At first I use matplotlib to plot the 2D normal distribution.



Calculating and Visualizing The Score Function

1. Probability Density Function (PDF): For a bivariate Gaussian distribution with mean $= \mu$ and covariance matrix $= \Sigma$ we have:

$$p(\mathbf{x}) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$

2. Calculating The Potensial Function: First we calculate the potensial function which equals the log of PDF function.

$$\log p(\mathbf{x}) = \underbrace{-\log(2\pi)}_{\text{constant}} - \frac{1}{2} \log |\Sigma| - \frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)$$

3. Score Function Calculation: The score function is equal to the deravitive of potensial function $\log p(\mathbf{x})$ with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\Sigma^{-1}(\mathbf{x} - \mu)$$

This derivation need linear algebra to calculate.

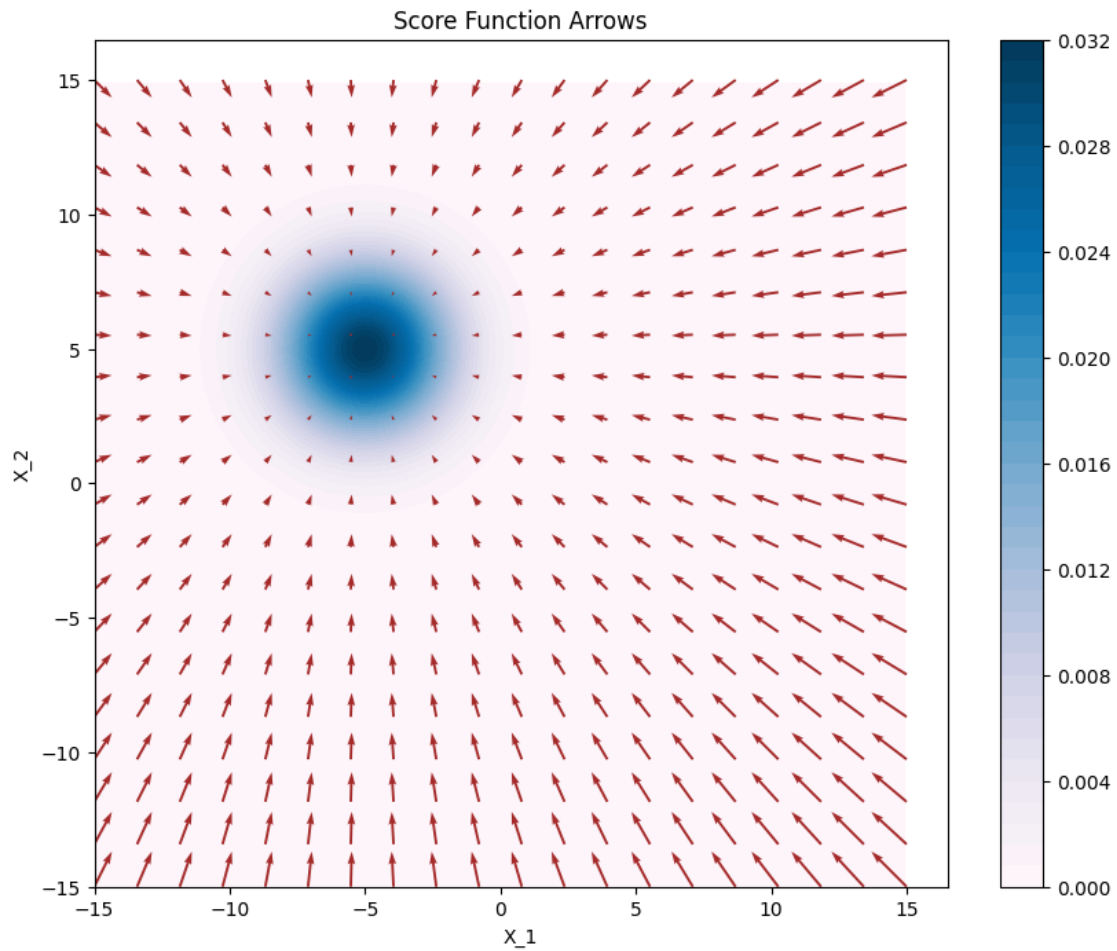
4. In This Case: Because the matrix is symmetrical:

if $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$, then:

$$\Sigma^{-1} = \begin{bmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{bmatrix}$$

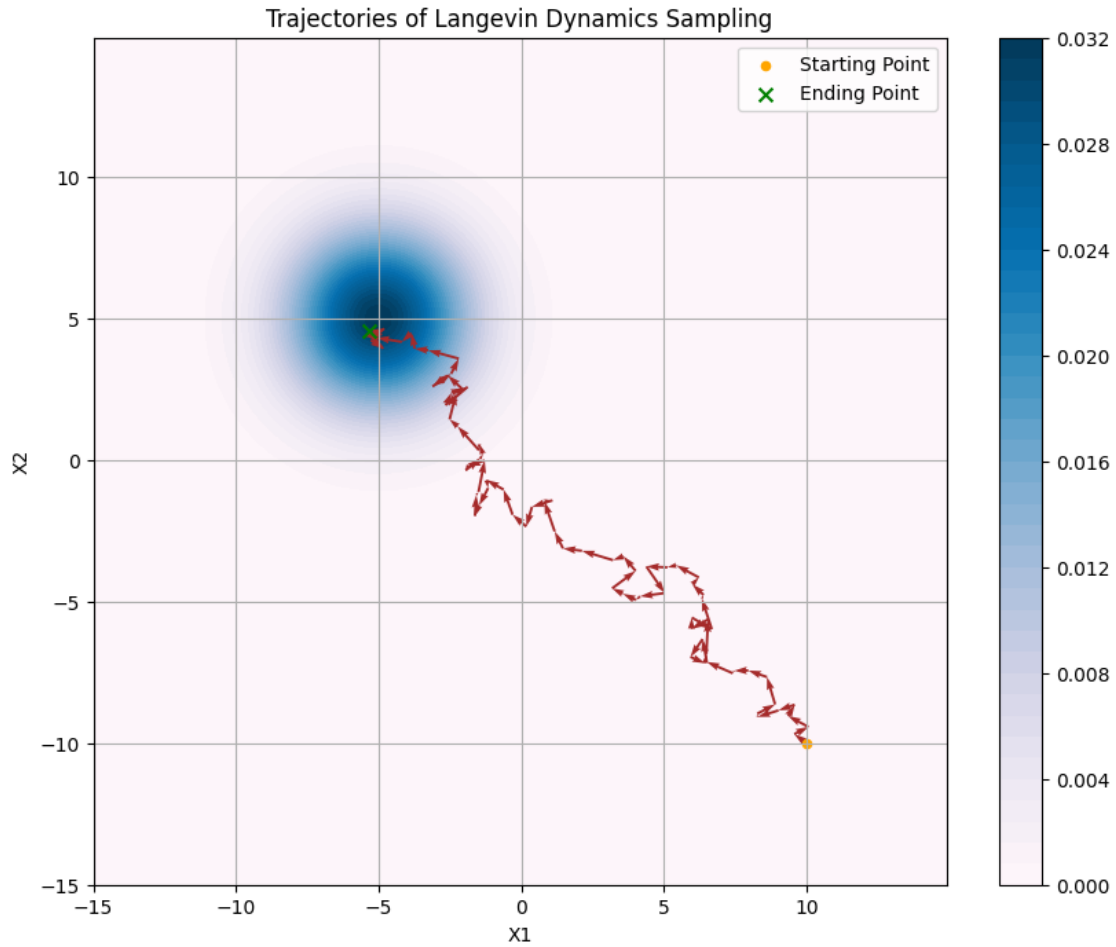
So the score function will be:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = S(\mathbf{x}) = - \begin{bmatrix} \frac{x_1 - \mu_1}{\sigma_1^2} \\ \frac{x_2 - \mu_2}{\sigma_2^2} \end{bmatrix}$$



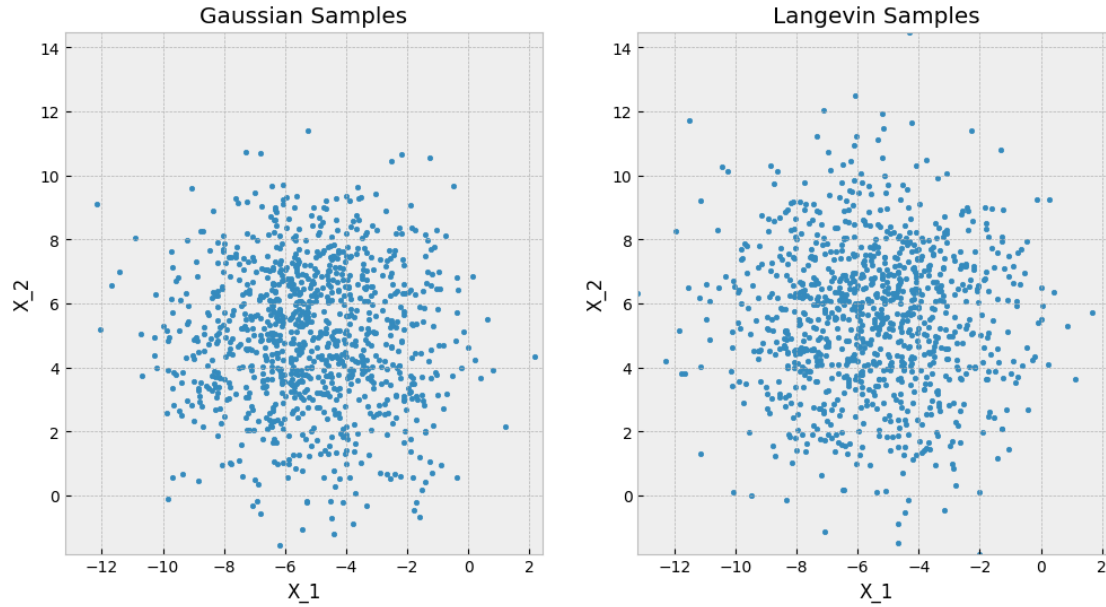
Now we implement the Langevin Dynamics algorithm:

Now let's run the algorithm and see the trajectory of the sampling!



Evaluating The Produced Sampling

Now we do a sampling on both the real gaussian distribution and the produced distribution. Then we'll compare them and evaluate the algorithm.



Other Ways To Compare The Distributions

To compare the distribution of the samples there are some other ways than visualization.

One way is to compare the statistics of the both distributions. As we can see below the cell code, the moments of the functions are approximately the same.

Langevin	Gaussian

Mean X: -5.5285	-5.1739
Var X: 5.6483	5.0821
Mean Y: 5.5391	4.9717
Var Y: 5.3109	4.9173
Covariance:	
[[5.65391992 -0.07539778]	
[-0.07539778 5.31623585]]	
vs	
[[5.08714604 -0.13171011]	
[-0.13171011 4.92220492]]	

0.0.1 Comparison Matrics

There are some measures that similarity of them might cause the similarity of the distributions. I've used two of them here. As we can see, both the distances are close considerably.

Dimension X:
 Energy Distance: 0.1907
 Wasserstein Distance: 0.3649
 Dimension Y:

Energy Distance: 0.2829

Wasserstein Distance: 0.5686

0.0.2 Multivariate Kolmogorov-Smirnov Statistical Test

We can also use Kolmogorov-Smirnov Statistical Test which shows how much two distributions are similar.

X1-axis KS statistic: 0.0910, p-value: 0.0005

X2-axis KS statistic: 0.1180, p-value: 0.0000

Bonus Question

The general answer is NO. But under some conditions we might be able to.

If the two normal distributions had far means, it'll cause the mixture of them to have two modes. This will make a bad problem. Because the gradient of $\log(p(x))$ is so close to 0 at a mode of function, when the algorithm reaches a peak it'll get stuck in it. This may stop the algorithm from progressing and discover the other parts of the distribution and make the sampling invalid.

If the $p(x)$ function had two close modes it would be probable for the langevin dynamics algorithm to be efficient and accurate. But we can't guarantee that. So I answer this question with "No."